PS000002: CNG SmartPlan

Simple Project Context for Seva First Innovation Challenge 2026

Official problem statement: Station-Level Demand Forecasting for CNG Sales to Optimize Network Expansion

Industry / department: Gujarat Gas Company Limited (as shown in the problem statement)

Purpose of this document: Clearly explain the problem, real-world examples, how our proposed solution addresses it, and the exact features planned for the project.

The project in one sentence

Build a software platform that estimates future CNG demand at individual stations, flags possible supply shortfalls, and helps evaluate where a new station may serve additional demand without unnecessarily taking customers away from nearby stations.

1. Understanding the Problem Statement

CNG demand is not the same at every station and can change over time. Demand depends on the number and type of vehicles nearby, traffic and travel patterns, weekdays and holidays, fuel-price differences, new stations opening nearby, and longer-term changes such as electric-vehicle adoption.

A company operating many stations needs to predict how much CNG each station may sell. It also needs to decide how much supply to plan, which stations or areas need attention, and where a new station should be built. If these decisions rely mainly on past sales averages or informal judgement, the company may underestimate demand at busy stations or invest in locations that do not bring enough additional customers.

The challenge is therefore not only to forecast sales. It is to use those forecasts to support day-to-day supply planning and longer-term station-location decisions.

2. Existing Problems and Simple Examples

Problem A: Demand is difficult to predict

Example: A station sold 6,000 kg of CNG yesterday. Tomorrow may be busier because of a holiday or changing traffic patterns. Simply assuming that tomorrow's sales will also be 6,000 kg may lead to poor planning. A forecasting system uses historical sales and relevant factors to estimate future demand.

Problem B: Busy stations may face queues or shortages

Example: A station is expected to need 8,000 kg tomorrow, but the current supply plan covers only 6,000 kg. The possible shortfall is 2,000 kg. An early warning gives the operations team time to review deliveries or other arrangements before demand exceeds the plan.

Problem C: A new station may take customers from an existing station

Example: An existing station sells 8,000 kg per day. A new station opens nearby and sells 3,000 kg, while the old station's sales fall to 5,000 kg. Combined sales remain 8,000 kg. The new station may have mainly divided the existing customer base instead of creating much new demand. This is called demand cannibalization.

Problem D: Areas with unmet demand may be overlooked

Example: A growing residential or commercial area has many CNG vehicles but no convenient station nearby. A new station in that area could serve demand that is currently difficult to reach. Location planning should consider demand, access and nearby stations—not just distance alone.

Problem E: Demand changes by hour, season and over years

Example: A station may be especially busy during commuting hours or festival travel. Over several years, CNG demand may also change as vehicle numbers and EV adoption change. Daily averages alone may miss these patterns.

Note: All numerical sales figures in this document are hypothetical examples for explanation. They are not actual Gujarat Gas data.

3. Our Proposed Solution

We propose CNG SmartPlan: a software-based demand forecasting and station-planning platform. It will combine sales forecasting with operational alerts and a tool for evaluating proposed new station locations.

The platform is intended to help users answer three questions:

- How much CNG might each station need in the coming days or weeks?
  - Which stations may have a mismatch between predicted demand and planned supply?
  - If a new station is proposed, could it serve additional demand, or might it mainly shift customers from nearby stations?

4. Exact Features We Plan to Build

Feature 1: Station-level CNG demand forecasting

Predict daily or weekly sales for each station using available historical sales and relevant factors. Show historical sales alongside the forecast so users can understand the trend. The forecast must show uncertainty or limitations when the data is incomplete.

Feature 2: Potential shortage and peak-demand alerts

Compare predicted demand with planned supply, where supply data is available. Flag stations where demand may exceed the plan. Highlight expected busy periods if hourly or time-of-day data is available. The alert supports human review; it does not itself deliver gas or guarantee that queues will disappear.

Feature 3: Map and ranking of candidate station locations

Display existing stations on a map and let the user assess proposed locations. Rank candidate areas using available indicators such as estimated demand, proximity to existing stations, accessibility and possible service gaps. The ranking should explain the main reasons behind each recommendation.

Feature 4: New-station impact simulator (our proposed signature feature)

Let the user choose a candidate location and compare an estimated 'before' and 'after' situation. Estimate demand the new station might attract, potential sales reductions at nearby stations, the change in combined network demand served, and whether the area has unmet demand. This feature directly addresses the problem statement's concern that new stations may cannibalize nearby stations. It must be presented as an estimate, not a guarantee.

Feature 5: Forecast reliability indicator

Show whether a prediction is supported by sufficient relevant data and how uncertain it is. A station with good historical records may have a more dependable forecast than a proposed station with no sales history. Clearly label synthetic-data demonstrations and avoid claiming real-world accuracy until tested on real data.

Feature 6: Long-term demand scenarios (if time permits)

Allow users to compare possible future outcomes under different assumptions about CNG vehicle growth, EV adoption, fuel-price differences and new station openings. This supports longer-term planning; it should not present scenario assumptions as certain predictions.

5. How Each Feature Solves the Problem

Existing problem	Feature that addresses it	Expected help
Future sales are uncertain	1. Demand forecasting	Better-informed daily and weekly planning
Demand may exceed planned supply	2. Shortage / peak alerts	Earlier review of supply and operations
New stations may overlap with existing ones	3. Location map and ranking 4. Impact simulator	Compare candidate sites and estimate cannibalization
Forecasts may be unreliable when data is limited	5. Reliability indicator	Make uncertainty and data gaps visible
Demand may change over several years	6. Scenario planner	Compare alternative long-term assumptions

6. Example: How the Whole System Works

Imagine the company is considering a new station in Area X. CNG SmartPlan first forecasts demand for nearby existing stations. It flags any station whose predicted demand may exceed planned supply. It then compares Area X with other candidate locations using estimated demand, accessibility and proximity to existing stations. The impact simulator estimates how much demand the new station might attract and how much nearby stations might lose. Finally, the reliability indicator explains which estimates are well supported by data and which need further investigation.

The result is a decision-support tool: it gives managers evidence to consider before changing supply plans or committing money to a new station. It does not automatically make decisions or guarantee business results.

7. Data and Prototype Assumptions

The most useful data would be historical station-level sales (daily or hourly), existing station coordinates, planned supply or deliveries, vehicle-registration trends, fuel prices, calendar and holiday information, and relevant geographic indicators. Some public location and vehicle data may be available, but detailed station-level sales and queue data may be proprietary; their availability has not been confirmed.

If company data is not provided, the team can build a demonstration using publicly available information where appropriate and clearly labelled synthetic sales data. Synthetic data can demonstrate the software workflow, but it cannot prove the model predicts real CNG demand accurately. Real-world performance must be validated using authorized real data when available.

8. What Makes Our Proposed Solution Meaningful?

Demand forecasting by itself is a common idea. Our intended emphasis is connecting forecasts to investment decisions. The new-station impact simulator is the main differentiator we plan to demonstrate: rather than only marking a location on a map, it estimates the possible effect of opening a station on nearby stations and the wider network. The reliability indicator also helps prevent users from treating uncertain estimates as facts.

This combination is a proposed design, not a claim that no similar system exists. The team should avoid claiming that the solution is completely unique until existing products and systems have been investigated.

9. Final Project Definition

Project name: CNG SmartPlan

Problem statement ID: PS000002

Official title: Station-Level Demand Forecasting for CNG Sales to Optimize Network Expansion

Target users: CNG demand-planning, sales and marketing teams; network operations and supply-planning teams; and CNG station franchisees/dealers evaluating local demand and station investment.

Why we are building it: To help users forecast station demand, identify possible supply mismatches, and evaluate new station locations while considering demand overlap and uncertainty.

Core features to prioritize: (1) station-level demand forecasting, (2) shortage and peak-demand alerts, (3) candidate-location map and ranking, (4) new-station impact simulator, and (5) forecast reliability indicator. Add long-term scenario planning if time and data allow.

Signature feature: New-station impact simulator that estimates incremental demand and possible cannibalization before an investment decision.

Important limitation: The usefulness and accuracy of forecasts depend on the quality and quantity of data. The prototype must distinguish actual data from synthetic data and must not claim unverified accuracy.

This document is the concise context for understanding the problem and the proposed solution. It focuses only on the problem, examples, solution logic and planned features.